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# Palaeomagnetic study of the Derbyshire lavas and intrusions, central England: definition of Carboniferous apparent polar wander

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ABSTRACT

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The remanence in extrusive volcanic rocks of Lower Carboniferous (Visean) age in the Pennines of central England has been reinvestigated, employing detailed thermal demagnetisation and incorporating new geological knowledge of field relationships. Indications of primary remanence identified by early studies are supported and four successive reversals in the Asbian–Brigantian succession (about 335 Ma) recognised. A mean (reversed) direction of $D/I = 203/36°$ (10 sites, $R = 9.75$) yields a pole position at 336°E 14°S ($dp/dm = 6/10°$). Grouping improves slightly when correction is made for mild intra-Carboniferous folding and widespread uniformity of polarity is recognised within individual units. Sporadic Variscan overprinting increases in importance to the southeast. Intrusive dolerite bodies yield two contrasting directions of magnetisation: in the central part of the Derbyshire pericline mixed polarity directions (mean $D/I = 211/37°$ (4 sites, $R = 3.92$)) comparable to the lavas imply a Lower Carboniferous age of emplacement; in the SE Upper Permian (?) directions of reversed polarity (mean $D/I = 221/-23°$) define a later intrusive event or comprehensive overprinting.

The primary Lower Carboniferous pole positions accord with contemporaneous poles from the Midland Valley of Scotland and demonstrate that Carboniferous and younger tectonic movements are palaeomagnetically undetectable, at least between these two areas. Apparent polar wander (APW) movement is confirmed between Lower and Upper Carboniferous times at about 335–320 Ma. The Lower Carboniferous limestones have a remanence entirely younger than depositional ages and are therefore unreliable indicators of Carboniferous magnetostratigraphy and tectonics. Silicified limestones prove to have remanences closer to the primary Lower Carboniferous field and appear to have been shielded from the effects of later overprinting. The APW swathe identified from North American Carboniferous rocks coincides in polarity and position with the latter part of the British Carboniferous path on the Pangaean reconstruction.

## 1. Introduction

The most prominent feature of the European palaeomagnetic record from Carboniferous rocks is a compact group of south poles plotting in the central-south Atlantic near to 340°E, 40°S (Piper, 1987). These poles are derived mostly from reversed polarity magnetisations. Much direct and circumstantial evidence shows that they record a prominent overprinting event during late Carboniferous and Permian times which also dominates much of the older Palaeozoic data (Piper and Grant, 1989). A small number of studies from Lower Carboniferous rocks yield magnetisations with mixed polarities and poles removed from this group. Early stepwise thermal demagnetisation and analysis of magnetostratigraphy in the Kinghorn lavas from central Scotland (Wilson and Everitt,

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1963) implied that the Lower Carboniferous pole position actually lay near 330°C, 10°S. This result has been stubstantiated by later study (Torsvik et al., 1989) and if confirmed elsewhere will imply that Carboniferous times were characterised by appreciable apparent polar wander (APW) movement.

A suite of Carboniferous rocks which have featured in studies since the earliest days of palaeomagnetism are the volcanics of the Derbyshire Dome in the Pennine Hills of central England. The results are summarised in Table 1. They include investigations (Belshé, 1957; Clegg et al., 1957) made before cleaning methods were routinely employed. Subsequent studies (Everitt and Belshé, 1960) found evidence for primary remanence and ancient field reversal. The axis of remanence was oriented NE − /SW + contrasting with the typical southerly shallow-directed remanences in most other European Carboniferous rocks. Later unpublished studies include more extensive sampling and alternating field (a.f.) demagnetisation (Titman, 1971; Moyse, 1973) and thermal demagnetisation (Titman, 1971). They also recognised reversals of magnetisation and directions divergent from remanence in contemporaneous limestones in northern England (Turner and Tarling, 1975; Turner et al., 1979).

Unfortunately, both Titman and Moyse experienced difficulties in differentiating between lavas and intrusions and were not able to take advantage of recent geological investigations (Institute of Geological Sciences, 1969, 1970, 1975a,b, 1976a,b, 1977a,b; Walters and Ineson, 1981) clarifying the nature and stratigraphic position of outcrops in these poorly exposed rocks. The present study has been undertaken to exploit this knowledge and resolve the nature of magnetic remanence by employing modern techniques of stepwise thermal demagnetisation and component analysis. It supplements study of six units in this area by Rolph (1984) published in Rolph and Shaw (1985) and undertaken to derive estimates of the palaeointensity of the Lower Carboniferous geomagnetic field.

## 2. Geological outline

The lavas are interleaved with marine shelf limestones belonging to the Tournasian and Visean epochs of the Dinantian Subsystem of the Silesian System. The limestones attain a total thickness of 660 m in this area (Cope, 1973; Dunham, 1973). They are succeeeded by a deltaic infill to the marine basin (the Millstone Grit Series) assigned

TABLE 1

Summary of previous palaeomagnetic investigations of the Derbyshire Volcanics

| No. of sites | No. of samples | $D$ | $I$ | $\alpha_{95}$ | $k$ | $R$ | Notes |
|---|---|---|---|---|---|---|---|
| *Study: Clegg et al. (1957)* | | | | | | | |
| – | 5 | 218 | 36 | – | – | – | Unspecified rocks baked by intrusion at Tideswell Dale |
| *Study: Belshé (1957)* | | | | | | | |
| – | 9 | 48 | 47 | 13 | – | – | |
| *Study: Everitt and Belshé (1960)* | | | | | | | |
| 1($N$) | 3 | 48 | −45 | – | – | – | Lavas and baked contact. |
| 3($R$) | 16 | 200 −221 | 29 −36 | | | | Positive contact test and reversal identified |
| *Study: Titman (1971)* | | | | | | | |
| 3 [a] | 6 | 23.4 | −60.7 | 9.9 | 156.1 | 2.9 | A.f. and thermal demag- |
| 10 [b] | 25 | 198.7 | 40.1 | 7.4 | 43.6 | 9.8 | netisation |
| *Study: Moyse (1973)* | | | | | | | |
| 5 | 29 | 41.8 | −40.9 | 5.5 | 133 | 4.97 | A.f. demagnetisation |

[a] Calculated from 3 sites in the Shacklow Wood Lava.

[b] Calculated from 10 reversed sites in lavas and intrusions, Waterswallows and Tideswelldale.

to the Namurian Subsystem. Faulting and folding occurred between Dinantian and Namurian times and also during the Variscan orogenic episode in Late Carboniferous and Permian times. Collectively, these movements folded the rocks into a pericline (the Derbyshire Dome). The Dinantian rocks are exposed in the core and dip outwards at shallow angles (Fig. 1).

Pillow lavas recorded in boreholes, and bedded hyaloclastite tuffs are evidence of submarine eruption. Deposition of the Dinantian sediments took place in an environment of shallow and fluctuating sea levels (Walkden, 1977) and the major lava units are often of great lateral extent and evidently the products of subareal volcanism. They are mapped for lateral distances of up to 14 km (Fig. 1). Over 30 lava and tuff units of olivine basalt composition have been recognised. Some are known only in mineworkings and boreholes; in other cases lavas merge laterally into tuffs (Walters and Ineson, 1981). Extrusive activity ranged over about 20 Ma from Upper Holkerian to Upper Brigantian times ($S_2$ to $D_2$ zones) with the major part occuring during Lower Brigantian times (Fig. 1). The geographic distribution indicates four centres of activity: a centre in the SE produced the Matlock lavas. An (Alport) centre further north was responsible for the Lees Bottom, Shacklow Wood, Lathkill Lodge and Conksbury Bridge lavas, while a centre in the west produced the

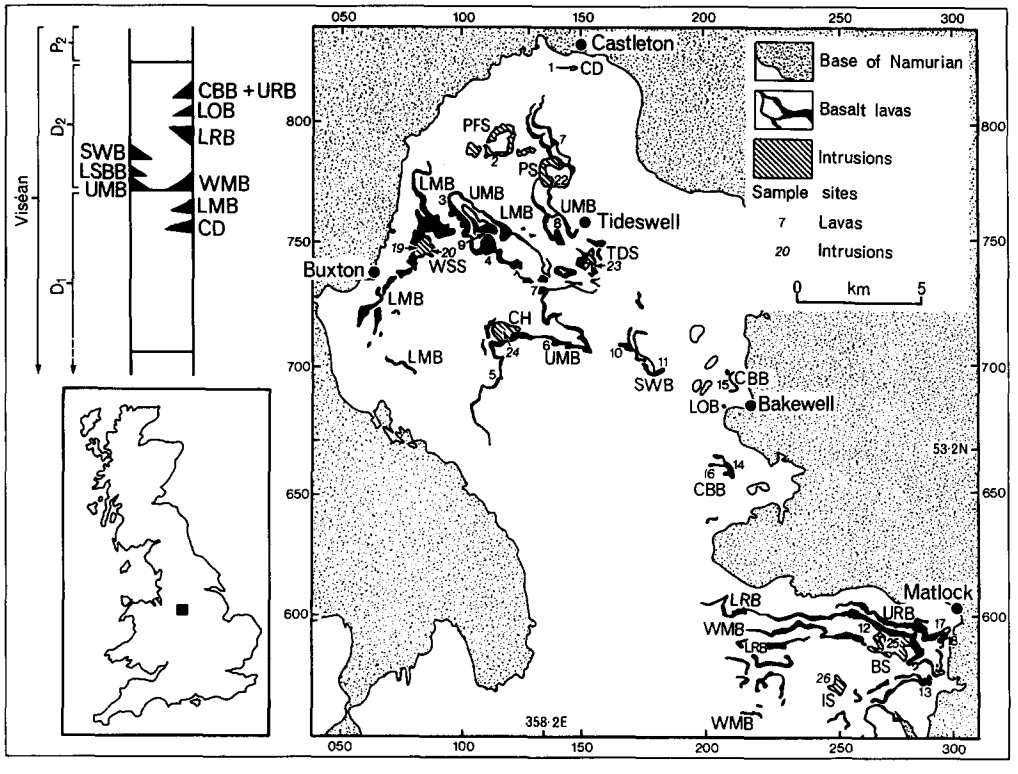



Fig. 1. The distribution of lavas and intrusions within the Lower Carboniferous (Dinantian) outcrop of the Southern Pennines in central England and the palaeomagnetic sampling sites of this study. Certain pyroclastic volcanic vents not sampled in this study are omitted for clarity. The named volcanic units are abbreviated as follows: UMB, Upper Miller's Dale lava; LMB, Lower Miller's Dale lava; CBB, Conksbury Bridge lava; LOB, Lathkill Lodge lava; LRB, Lower Matlock lava; WMB, Wintersmoor lava; URB, Upper Matlock lava; PFS, Peak Forest sill; PS, Potluck sill; WSS, Water Swallows intrusion; TDS, Tideswelldale intrusion; BS, Bonsall intrusion; IS, Ible intrusion. The inset figures show the regional location of this outcrop and the stratigraphic position of the lava units within the Lower Carboniferous stratigraphy; the latter comprises mostly marine carbonates although pyroclastic sequences are locally important. For further details of sampled units see Table 2.

TABLE 2

Location and description of sample sites

| Site no. | Name of rock unit | Grid reference | Cores sampled | Notes |
|---|---|---|---|---|
| *(i) Lower Carboniferous (Dinantian) lavas* | | | | |
| 1 | Cave Dale | 41483822 | 11 | |
| 2 | Lower Miller's Dale | 41453781 | 14 | (Also near sites PL1 and 2 of Rolph (1984)). |
| 3 | | 40963764 | 10 | This unit comprises up to 3 |
| 4 | | 41093748 | 10 | extrusive events in Asbian limestones (former $D_1$ zone) |
| 5 | Upper Miller's Dale | 41163698 | 13 | This unit located at |
| 6 | | 41433709 | 9 | Asbian–Brigantian boundary ($D_2/D_1$ zones) |
| 7 | | 41343736 | 12 | |
| 8 | | 41433769 | 10 | |
| 9 | | 41063754 | 13 | (Also site BH1 of Rolph (1984)). |
| 10 | Lees Bottom | 41693705 | 12 | These units formerly equated with unit above; now regarded as of younger (Brigantian) age |
| 11 | Shacklow Wood | 41813698 | 18 | |
| 12 | Lower Matlock | 42573599 | 10 | Up to three distinct flows; merges laterally into a tuff |
| 13 | | 42833575 | 11 | Also site BW1 of Rolph (1984) |
| 14 | Lathkill Lodge | 42033662 | 14 | |
| 15 | Conksbury Bridge | 42123690 | 19 | Same extrusive event as Alport lava |
| 16 | | 42113661 | 17 | |
| 17 | Upper Matlock | 42903590 | 10 | |
| 18 | | 42903590 | 6 | (Also site ML1 of Rolph (1984)) |
| *(ii) Post-Visean intrusive bodies* | | | | |
| 19 | Waterswallows | 40833748 | 13 | Sites at shallow and deep |
| 20 | | 40833748 | 12 | levels of quarry cutting. Also site WS1 of Rolph (1984). Dated $311 \pm 6$ Ma (K–Ar) |
| 21 | Peak Forest | 41163788 | 11 | Also site DF1 of Rolph (1984) |
| 22 | Potluck | 41413783 | 7 | |
| 23 | Tideswelldale | 41523740 | 16 | Emplaced into Lower Miller's Dale lava. Dated $287 \pm 13$ Ma (K–Ar) |
| 24 | Calton Hill | 41203715 | 11 | Olivine dolerite with peridotite nodules emplaced into a vent cutting the Upper Miller's Dale lava. Dated $295 \pm 14$ Ma (K–Ar) |
| 25 | Bonsall | 42753590 | 13 | Probably largest igneous intrusion in S. Pennines. Emplaced into crest of Matlock anticline |
| 26 | Ible | 4253570 | 10 | Olivine dolerite sill subject to strongest low grade alteration in this area; widespread chloritisation. Dated $309 \pm 21$ Ma (K–Ar) |

Miller's Dale lava (Fig. 1); a fourth (Longstone Edge) centre is not significantly represented by surface outcrop. Fitch et al., (1970) report K–Ar ages of 282 ± 17 Ma for the Lower Miller's Dale lava and 318 ± 9 Ma for the Upper Miller's Dale lava. In common with some estimates from the intrusions (see below and Table 2), Fitch et al. considered these dates on the low side owing to argon loss.

The intrusive rocks comprise a small number of olivine dolerite and ultrabasic bodies of sill-like form (Fig. 1). Potassium–argon determinations range from 311 to 287 Ma (Fitch and Miller, 1964; Stevenson et al., 1970) but error limits overlap (Table 2) and they could be part of the same suite. Critical observational data are summarised in Table 2.

Hydrothermal mineralisation is concentrated on joints and fault fissures in the limestones but is locally found influencing the lavas. It appears to have taken place predominantly in Permian times but probably continued into the Mesozoic era (Ineson and Mitchell, 1973). Mild Hercynian folding reflects the large distance of this region from the Hercynian orogenic front to the south and provides only limited scope for the palaeomagnetic fold test. Younger structural deformation is also minimal and related mainly to rifting associated with Mesozoic sedimentary basins to the east and west.

## 3. Palaeomagnetic studies

### *3.1. Sampling and treatment*

The present investigation has aimed to sample and study each lava unit exposed as surface outcrop. Exposures are invariably too poor to identify whether they represent single or multiple flows. Walters and Ineson (1981) show from borehole evidence that many lava units comprise several flows. We follow tradition by referring to them as 'lava' and use the wide geographic extent of the Lower Miller's Dale and Upper Miller's Dale lavas to test the coherency of the remanence record by sampling at 3 and 5 distributed sites, respectively (Fig. 1 and Table 2). Altogether the lava stratigraphy was sampled at 18 sites. Only the Wintersmoor lava in the mapped units (Institute of Geological Sciences, 1969–1977 and Fig. 1) provided no suitable outcrop for palaeomagnetic sampling. The intrusions have been sampled at seven sites (Table 2), one on each of the exposed bodies.

Up to 18 separate cores were drilled at each site with portable two-stroke drills and oriented in situ by sights on the Sun and magnetic bearings linked to topographic sites. In the laboratory cores were sliced into 2.4 cm long cylinders and measured by 'Minispin' magnetometers. Although a.f. demagnetisation was undertaken, this method only properly recovered the component structure when a viscous overprint was the single addition to the characteristic remanence; at many sites a history of partial ancient secondary overprinting could only be recognised by thermal demagnetisation. Accordingly, only this method of treatment was pursued and all results discussed here are based on it. Steps of 100°C to 300°C, 100°C or 50°C to 500°C, and then in 20°C steps to 600°C were employed, or higher if a coherent component remained at this stage. Susceptibility measurements were used to monitor changes in the magnetic mineralogy during progressive cleaning. Following initial study of directional stereoplots, the component structure was resolved by study of orthogonal vector plots and principal component analysis.

### *3.2. Results from the lava succession*

The lowest (Cave Dale) lava is dominated by a coherent single component of distributed blocking temperature spectrum (Fig. 2). This component is of reversed polarity with the maximum unblocking temperatures indicative of low-Ti carriers. Sites 2, 3 and 4 in the Lower Miller's Dale division also have a reversed characteristic magnetisation. A south to southwest (S–SW) positive component is the dominant feature of these sites; in some samples at site 2 (eg. sample 2–4 in Fig. 2) this is a single component subtracted by 600°C; more commonly a steeper positive component is first subtracted at low temperatures (eg. sample 4–1 in Fig. 2). A feature peculiar to some cores from site 2 is a large NW-directed component subtracted at low temperatures to recover an easterly-directed

component. In one sample (2–15 in Fig. 2) the former is discrete and subtracted below 200°C. It may therefore reside in goethite. In others the component overlaps with the higher blocking temperature component which remains above the Curie point of magnetite but is poorly defined ($D/I = 78/-17°$, 4 samples, $R = 3.64$); it corresponds to no known younger field direction for this area and we suspect that continuous conversion of goethite to hematite is taking place during demagnetisation. Cores exhibiting this behaviour were adjacent to a mineral vein; the remainder located more than 4 m away from the vein have properties similar to other sites in this division (2(i) in Table 3).

A characteristic magnetisation of normal polarity is recognised at three of five sites in the Upper Miller's Dale flow. A NNE–NE negative remanence (eg. samples 5–8 in Fig. 2) is typical of these sites. It is essentially resident in magnetite; a small residual component with the same direction is recognised in the hematite range (580–680°C) in some samples and is therefore probably of deuteric origin. At site 7 this is not the sole component of the remanence (eg. samples 7–4 in Fig. 2); above 520°C directions tend to move into a SW negative quadrant but intensity behaviour is then erratic and no coherent components are defined. Site 8 is from a poorly exposed and strongly exfoliated outcrop where magnetisations are an order weaker than elsewhere. With the exception of two more strongly magnetised samples yielding components similar to sites 5–7 near the Curie point of magnetite, cores here are dominated by a SW negative remanence (Table 3 and samples 8–2 in Fig. 2) unblocked below 400°C and therefore likely to be of secondary origin. Site 9 has a remanence contrasting with the remaining sites in this division and dominated by a northerly steep negative component fully subtracted by 520°C; magnetisations have comparable magnitudes to the remaining sites and there is no clear evidence to link this site to a geomagnetic field transition. The succeeding Lees Bottom Flow has reversed magnetisation supporting geological evidence that it is not correlative with the Upper Miller's Dale lava (Walters and Ineson, 1981). All cores are dominated by a convergent SW positive component (eg. sample 10–9 in Fig. 3) following removal of a viscous component. The Shacklow Wood lava is again dominated by a single magnetite-resident component but in this case with a NE negative direction (eg. sample 11–5 in Fig. 3). The polarity is therefore normal although the inclination is notably steeper than the other units with this polarity except site 9. Outcrop is poor but interleaved limestones dip at little more than 10° and tilt adjustment has little effect on this direction (Table 3). There are therefore palaeomagnetic grounds (see site 9) for linking this lava with part of the Upper Miller's Dale unit with which it was formerly correlated (Walters and Ineson, 1981).

Two sites (12 and 13) from the succeeding Lower Matlock lava are both dominated by SW negative components following removal of a viscous positive remanence. At site 12, final subtraction of the convergent vector by 660°C is indicative of a common magnetite and hematite-held remanence (sample 12–10 in Fig. 3). There is no clear indication of the SW positive direction of reversed field polarity identified in the lavas below, although inclinations are distributed between $I = -25°$ and $I = 1°$; with no significant variation in declination, this smearing and the presence of hematite are suggestive of a secondary origin. The structure of remanence at site 13 is also complex. At temperatures above 500°C in most samples it veers towards a direction which is shallower and more westerly to produce the kink in the orthogonal plots (Fig. 3). The higher blocking temperature component survives to 660°C and is therefore, at least in part, hematite-held; both component means are listed in Table 3.

The Lathkill Lodge lava is very weakly magnetised (Table 3). Although eight components meet acceptability criteria, four yield a poorly defined NW + direction (eg. sample 14–13 in Fig. 3) comparable with some intermediate directions recognised elsewhere (Tables 3 and 4). It is possible that this flow erupted during a polarity transition. In the Conkesbury Bridge Unit dual component behaviour is observed at site 15. A S + component is subtracted below 300°C in some samples but survives close to the magnetite Curie point in others; the higher blocking temperature component has a similar declination but shallow inclina-

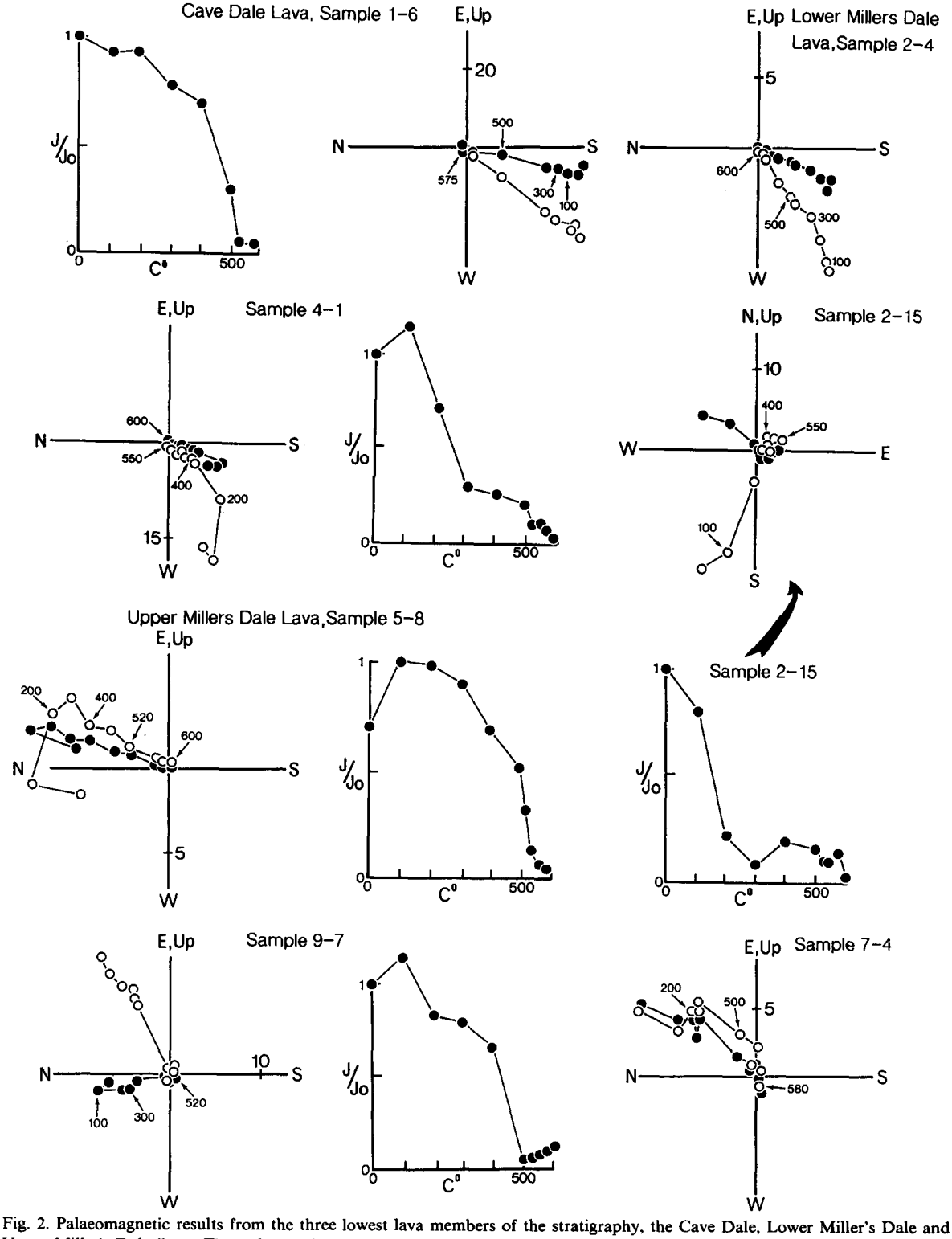



Fig. 2. Palaeomagnetic results from the three lowest lava members of the stratigraphy, the Cave Dale, Lower Miller's Dale and the Upper Miller's Dale flows. The orthogonal projections of the vectors onto the horizontal (solid symbols) and vertical (open symbols) planes illustrate subtraction of the remanence with progressive thermal cleaning (indicated in degrees centigrade). The axes are scaled in units of $\times 10^{-5}$ A.m$^2$ kg$^{-1}$. Reduction in magnitude of the total remanence is indicated as a fractional decrease of initial remanence, $J_0$.

tion and when this component predominates behaviour becomes erratic above 520°C. Site 16 in the same unit is dominated by the S + directed remanence subtracted by the Curie point of magnetite (sample 16–8 in Fig. 3).

At site 17 in the Upper Matlock unit a variable secondary component with steep positive direction is removed below 200°C to recover a shallow E-directed remanence retained to the Curie point of hematite (e.g. sample 17–3 in Fig. 3). The higher blocking temperature component ($D/I = 119/-19°$) is reminiscent of the direction recognised at site 7 lower in the stratigraphic column which does not correspond with any known younger field direction for this area. Site 18 yields a normal polarity component with positive in-

TABLE 3

Site mean palaeomagnetic results, Derbyshire (Lower Carboniferous) lavas, central England

| Site no. | $J_0$ ($\times 10^{-5}$ A $m^{-2}$ $kg^{-1}$) | $N/n$ | $D$ (in situ) | $I$ (in situ) | $R$ | $k$ | $\alpha_{95}$ | $D$ (Tilt adjusted) | $I$ (Tilt adjusted) | Polarity |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 18.4–3.8 | 11/9 | 196.0 | 31.5 | 8.87 | 61.5 | 6.7 | 192.6 | 27.9 | R |
| 2 | 43.1–10.6 | (i) 14/10 | 200.7 | 38.6 | 9.77 | 39.1 | 7.8 | 191.5 | 36.7 | R |
| | | (ii) 14/4 | 306.5 | 68.2 | 3.98 | 150.0 | 7.0 | 306.5 | 79.7 | N |
| 3 | 33.5–7.4 | 10/6 | 228.5 | 40.3 | 5.88 | 66.7 | 10.6 | 226.4 | 44.7 | R |
| 4 | 19.6–6.8 | 10/9 | 205.9 | 34.5 | 8.73 | 29.6 | 9.6 | 199.8 | 40.7 | R |
| 5 | 22.6–1.7 | 13/7 | 25.6 | −28.9 | 6.93 | 85.7 | 6.4 | 22.8 | −28.8 | N |
| 6 | 38.2–11.2 | 9/8 | 31.9 | −37.8 | 7.92 | 87.5 | 5.9 | 24.8 | −34.9 | N |
| 7 | 12.9–2.7 | (i) 12/6 | 43.7 | −33.6 | 5.84 | 31.3 | 12.1 | 36.9 | −36.3 | N |
| | | (ii) 12/6 | 123.9 | −14.4 | 5.87 | 38.5 | 10.8 | 124.5 | −24.3 | I |
| 8 | 1.8–0.01 | 10/7 | 302.4 | −19.8 | 6.81 | 31.4 | 10.9 | 207.0 | −19.8 | R |
| 9 | 20.7–12.4 | 13/12 | 354.8 | −62.8 | 10.59 | 26.8 | 9.3 | 352.8 | −70.8 | N |
| 10 | 9.6–0.6 | 9/9 | 229.1 | 50.0 | 8.83 | 47.0 | 7.6 | 220.5 | 52.6 | R |
| 11 | 37.1–0.6 | 10/8 | 21.6 | −67.1 | 7.87 | 52.5 | 7.7 | 1.7 | −63.1 | N |
| 12 | 45.3–1.0 | 11/9 | 228.8 | −11.8 | 8.85 | 52.3 | 7.2 | 227.9 | 3.9 | R |
| 13 | 157.7–3.2 | (i) 11/5 | 211.2 | −44.0 | 4.95 | 82.1 | 8.5 | 211.5 | −38.0 | R |
| | | (ii) 11/7 | 235.0 | −31.6 | 6.90 | 57.7 | 8.0 | 224.4 | −25.7 | R |
| 14 | 13.9–0.01 | 14/4 | 324.5 | 31.1 | 3.79 | 14.0 | 25.4 | 327.7 | 34.5 | I |
| 15 | 34.2–0.3 | (i) 9/6 | 193.0 | 19.6 | 5.95 | 98.7 | 6.8 | 190.6 | 28.0 | R |
| | | (ii) 9/7 | 187.0 | −11.8 | 6.74 | 23.4 | 12.7 | 187.8 | −3.9 | R |
| 16 | 52.8–1.5 | 18/6 | 195.8 | 17.7 | 15.73 | 55.0 | 5.0 | 194.9 | 22.1 | R |
| 17 | 77.3–0.5 | (i) 6/3 | 316.9 | 20.6 | 2.95 | 38.5 | 20.1 | 322.0 | 19.5 | I |
| | | (ii) 6/4 | 118.7 | −18.6 | 3.78 | 13.4 | 26.0 | 123.8 | −22.0 | I |
| 18 | 30.3–22.6 | 5/4 | 54.7 | 49.1 | 3.95 | 60.8 | 11.9 | 52.8 | 35.3 | N |
| | | | | | | | | Palaeomagnetic pole | | |
| | | | | | | | | °E | °S | $dp/dm$ |
| *Group mean of NE − /SW + axis* | | | | | | | | | | |
| 10 sites [a] (in situ) | | | 207.9 | 33.9 | 9.70 | 30.4 | 8.9 | | | |
| 10 sites [a] (tilt adjusted) | | | 203.0 | 35.8 | 9.75 | 35.6 | 8.2 | 335.9 | 14.3 | 5.5/9.5 |
| *Group mean of SW − magnetisations* | | | | | | | | | | |
| 5 sites [b] (tilt adjusted) | | | 211.6 | −17.2 | 4.69 | 12.7 | 22.3 | | | |
| 5 sites [b] (in situ) | | | 212.7 | −24.8 | 4.69 | 12.9 | 22.1 | 313.0 | 42.2 | 12.7/23.7 |

$D$ and $I$ are the mean declination and inclination derived from $n$ samples of a total site population of $N$ cores. $R$ is the resultant vector and $k$ is the Fisher precision parameter ( $= (n-1)/(n-R)$). $\alpha_{95}$ is the cone of 95% confidence about the mean direction. $dp$ and $dm$ are the radii of the oval of confidence about the palaeomagnetic pole in the colatitude direction and at right angles to it, respectively.

[a] Includes site means from 1, 2(i), 3, 4, 5, 6, 7(i), 10, 15 and 16.

[b] Includes site means from 8, 12, 13(i), 13(ii) and 15(ii).

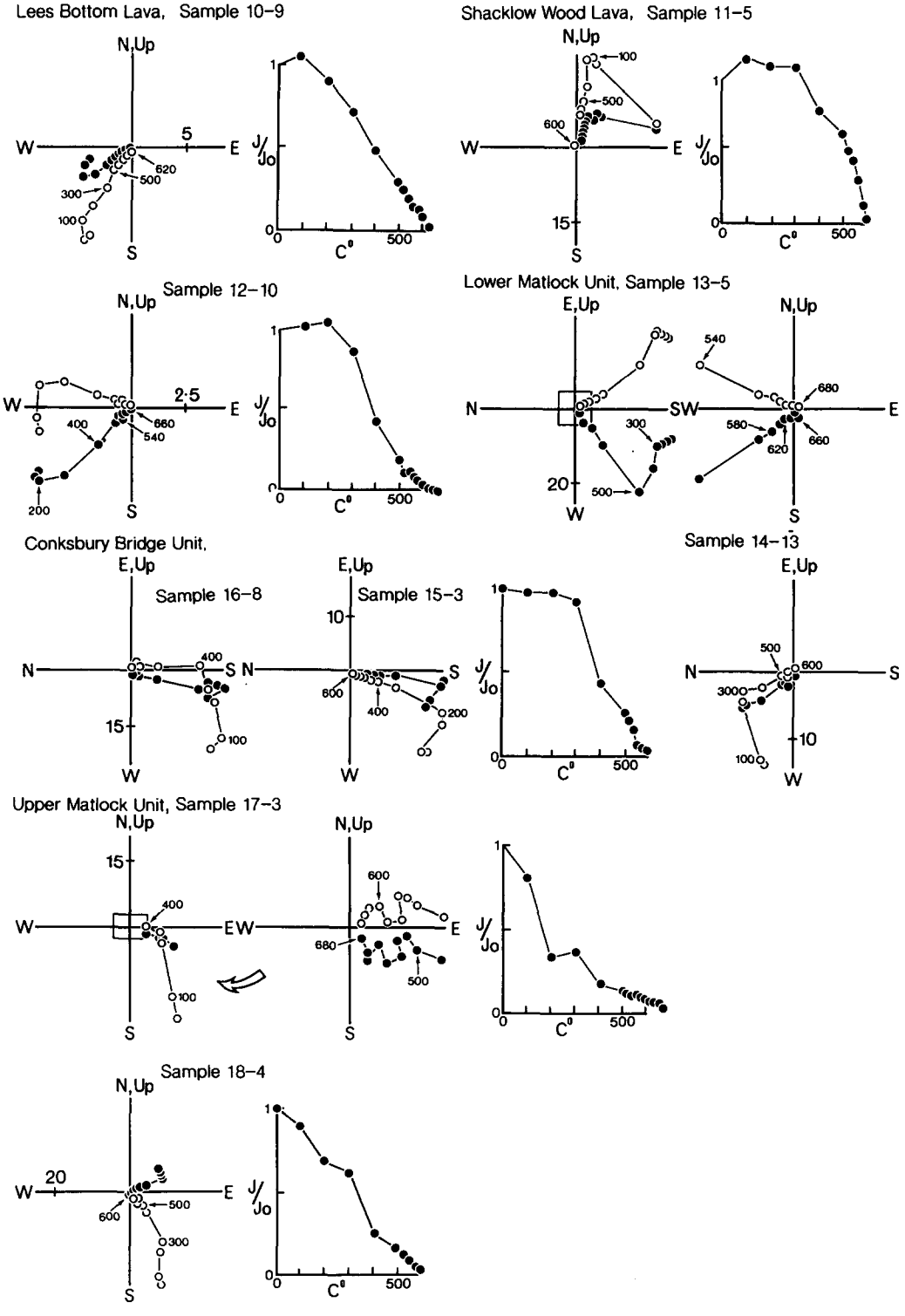



Fig. 3. Palaeomagnetic results from the higher (Asbian Stage) lavas in the Derbyshire succession. Numbers refer to the sampling sites of Fig. 1 and Table 2 and symbols are as for Fig. 2.

clination (eg. sample 18–4 in Fig. 3); this is only compatible with the regional field in Mesozoic times.

Site mean directions from the lavas are shown in Fig. 4. A dipolar axis is defined by NE directions (at sites 5, 6, 7 and possibly 11) and SW + directions (at sites 1, 2, 3, 4, 10, 15 and 16). These directions do not correspond to the predicted post-310 Ma field computed from the APW path for Eurasia (Piper, 1987 and Fig. 4); they are therefore inferred to predate these times and are probably primary. A group of SW directions of uniformly reversed polarity has similar declination but opposite inclination to the primary group and straddles the predicted palaeofield path at about 260 Ma. Since the lavas are undoubtedly of Lower Carboniferous age these directions are interpreted as Carboniferous–Permian overprints.

The subsidiary groupings of SE − and NW + directions should be noted (Table 3). Whilst they are suggestive of a dipolar axis, they do not correspond to the post-Carboniferous field. Owing to the low degrees of deformation there is no apparent tectonic explanation for these directions. It is possible that they record a prominent transitional field axis occupied during inversions of the field because: (i) they are in some cases significantly more weakly magnetised than samples defining the NE − /SW + axis; (ii) they are found at site 7 near an established reversal in the succes-

TABLE 4

Site and group mean palaeomagnetic results for dolerite intrusions, Derbyshire, England

| Site no. | $J_0$ ($\times 10^{-5}$A $m^{-2}$ $kg^{-1}$) | $N/n$ | $D$ (in situ) | $I$ (in situ) | $R$ | $k$ | $\alpha_{95}$ | $D$ Tilt adjusted | $I$ Tilt adjusted | Polarity |
|---|---|---|---|---|---|---|---|---|---|---|
| 19 (i) | 44.5– 5.2 | 13/11 | 189.5 | 34.3 | 10.96 | 235.1 | 2.9 | 194.7 | 32.6 | R |
| (ii) | | 13/3 | 202.7 | −9.3 | 2.96 | 53.6 | 17.0 | 201.3 | −12.3 | R |
| 20 (i) | 38.6–19.1 | 12/10 | 196.3 | 38.3 | 9.95 | 167.0 | 3.8 | 202.0 | 35.7 | R |
| (ii) | | 12/5 | 275.5 | 45.7 | 4.49 | 7.8 | 29.2 | 274.9 | 37.7 | I |
| (Waterswallows intrusion mean) | | 25/21 | 192.6 | 36.3 | 20.87 | 149.7 | 2.6 | 198.1 | 34.2 | R |
| 21 (Peak Forest) | 10.4– 2.1 | 11/10 | 343.9 | 68.7 | 9.63 | 224.5 | 9.9 | 343.4 | 65.7 | N |
| 22 (i) (Potluck intrusion) | 153.0–18.5 | 7/4 | 276.0 | 2.9 | 3.86 | 21.4 | 20.4 | 275.8 | 10.8 | I |
| (ii) (Potluck intrusion) | | 7/3 | 350.6 | −8.5 | 2.97 | 63.5 | 15.6 | 349.7 | −5.1 | N |
| 23 (Tideswelldale) | 37.2–27.2 | 16/15 | 222.6 | 34.2 | 14.38 | 22.4 | 8.3 | 217.3 | 37.6 | R |
| 24 (i) Calton Hill | 23.7–10.2 | 11/9 | 57.6 | −37.5 | 8.42 | 13.9 | 14.3 | 51.4 | −38.8 | N |
| (ii) | | 11/3 | 125.0 | −49.3 | 2.99 | 350.2 | 6.6 | 122.9 | −57.2 | I |
| 25 (Bonsall) | 16.2– 6.0 | 13/11 | 228.7 | −40.3 | 10.83 | 57.1 | 6.1 | 228.1 | −26.3 | R |
| 26 (Ible) | 25.2– 9.0 | 10/6 | 213.0 | −8.8 | 5.98 | 219.7 | 4.5 | 214.2 | −19.8 | R |
| | | | | | | | | Palaeomagnetic pole | | |
| | | | | | | | | °E | °S | $dp/dm$ |
| *Group mean results* | | | | | | | | | | |
| NE − /SW + axis [a] | 4 sites (in situ) | | 211.4 | 37.7 | 3.85 | 20.1 | 21.0 | | | |
| | 4 sites (tilt adjusted) | | 210.9 | 37.0 | 3.92 | 36.7 | 15.4 | 328.8 | 11.5 | 10.6/18.0 |
| Bonsall and Ible intrusions | | | | | | | | | | |
| | 2 sites (in situ) | | 219.8 | −24.8 | 1.91 | 11.1 | – | | | |
| | 2 sites (tilt adjusted) | | 220.9 | −23.2 | 1.98 | 64.0 | – | 304.2 | 37.6 | 18.0/33.8 |
| WNW/SSE axis [b] | 7 sites (tilt adjusted) | | 121.6 | −30.7 | 6.51 | 12.2 | 18.0 | 72.4 | 32.0 | 11.2/20.1 |
| | 7 sites (in situ) | | 120.3 | −27.1 | 6.50 | 12.0 | 18.2 | | | |

Symbols are as for Table 3. Palaeomagnetic poles are all quoted for tilt adjusted directions.

[a] Components 19(i), 20(i), 23 and 24(i).

[b] Components 7(ii), 14, 17(i), 17(ii), 20(ii), 22(i) and 24(ii).

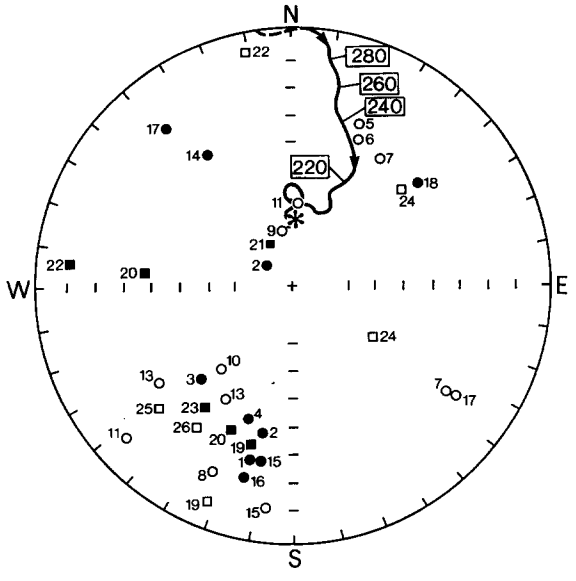



Fig. 4. Equal area projection showing site mean directions of magnetisation resolved by thermal demagnetisation of the Derbyshire lavas and intrusives by this study. Closed symbols are lower hemisphere plots and open symbols are upper hemisphere plots. Directions defining the NE−/SW+ axis are inferred to record primary remanences and are tilt adjusted, as are components showing the poorly defined WNW–ESE axis. Other directions are shown in situ. The arrowed line (dashed in upper hemisphere, solid in lower hemisphere) shows the palaeofield migration path for this area predicted from the mean APW path for Eurasia after 310 Ma (Piper, 1987).

sion; (iii) they are also identified in the intrusions (Section 3.3.) where they were apparently recorded by the more protracted cooling of these bodies.

Tectonic tilts are difficult to determine precisely and can seldom be evaluated at lava outcrop; in most cases local dips must be estimated from the more abundant outcrops of the adjoining limestones or from the regional strike contours. Tilts determined in this way are all less than 20° and are often as little as 5°; they generally reflect the periclinal structure of this district, although local divergencies may reflect small rotations near to ENE–WSW trending intra-Carboniferous faults. Directions adjusted for tilt are listed in Table 3. The essential features of the site mean distribution remain. Sites 11 and 12 are brought closer to the NE−/SW+ axis but are anomalously steep and shallow, respectively with respect to the other sites defining this axis; they are therefore excluded from the group mean (Table 3). The Fisher (1953) precision parameter, $k$, of the group mean defined by ten remaining sites defining this axis increases from 30.4 to 35.6 when adjustment for tilt is made. Because tilts are low this is appreciably less than the increase required to define significance at the 95 % confidence level ($k_2/k_1 > 2.22$ according to the palaeomagnetic fold test of McElhinny, 1964). However, it suggests that remanence defining this axis was acquired before intra-Carboniferous (post-Brigantian, pre-mid-Namurian, about 325 Ma) folding of these rocks. A primary origin for this remanence is strongly supported by the observation that the Lower Miller's Dale unit is consistently reversely magnetised (sites 2–4) while the Upper Miller's Dale unit is consistently normally magnetised (sites 5–7; see also earlier results noted in Section 3.3.), and evidence for a deuteric origin in some samples.

Lavas yielding SW-directed shallow negative components have more variable directions both before and after tilt correction and are therefore unlikely to record a single magnetising event. The calculated mean from 5 directions shows no appreciable change in precision when adjustment for tilt is made (Table 3).

### *3.3. Palaeomagnetic properties of the intrusive suite*

The intrusions are petrographically comparable with the lavas and have similar, but less variable, intensities of natural remanent magnetisations (c.f. Tables 3 and 4). Sites 19 and 20 come from deep and shallow levels, respectively in the heavily-quarried Waterswallows intrusion. The characteristic remanence (see also Rolph and Shaw (1985) site WS1) is a SW positive component fully subtracted by the Curie point of magnetite (eg. 19–1 in Fig. 5). A NW positive direction is observed as a low blocking temperature component in some samples from higher levels and composes the entire remanence in one sample (20–11, Fig. 5). It is present, but more sporadically so, in deeper levels of the body (see lower blocking temperature component in sample 19–1, Fig. 5). Higher levels exhibit visible hematisation along joint fractures not observed in the deeper site; although hematite

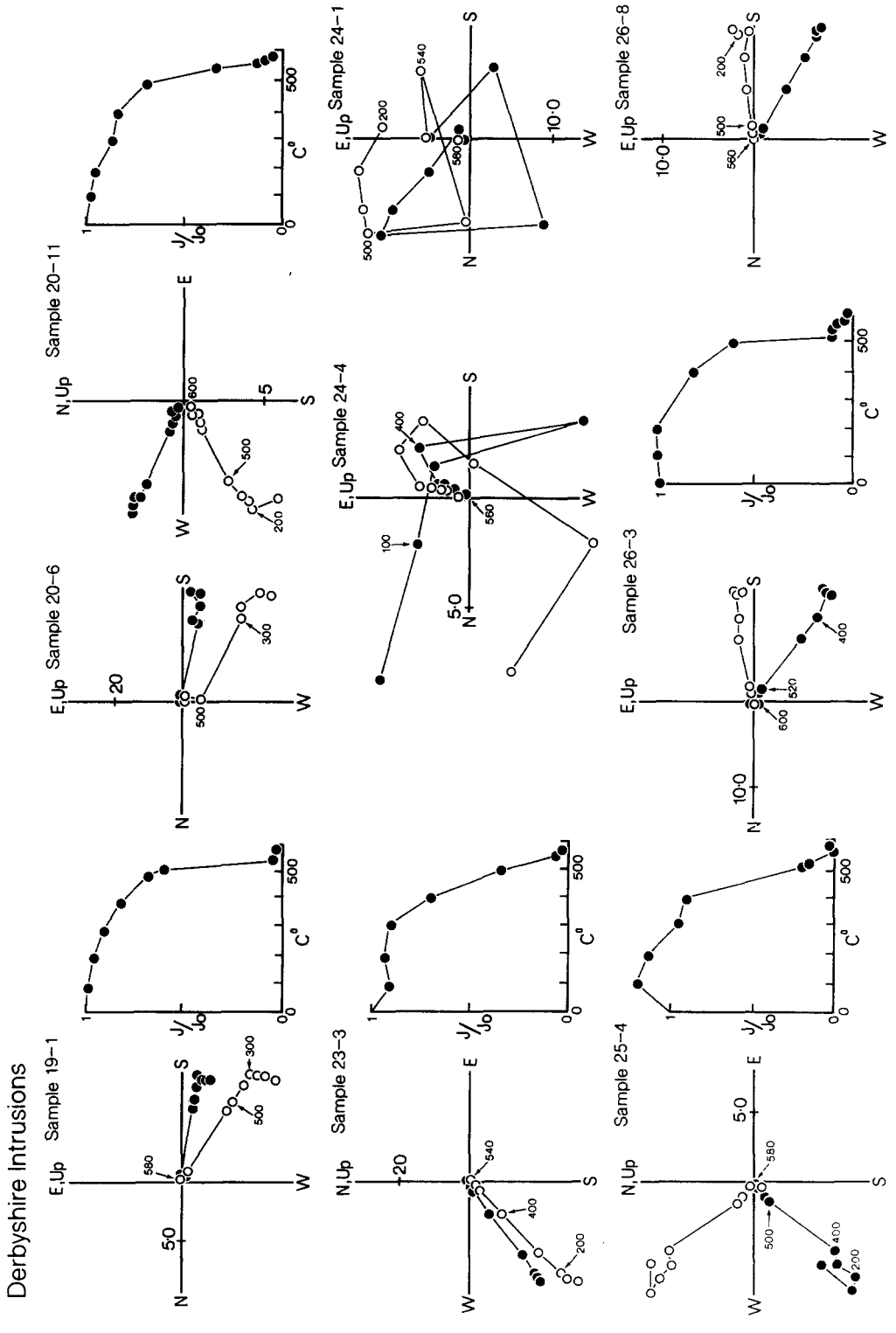



Fig. 5. Representative palaeomagnetic results from the dolerite intrusions in the Derbyshire volcanic district. Numbers refer to sampling sites of Fig. 1 and Table 2 and the symbols are as for Fig. 2.

has not been recognised as a magnetic carrier, it appears to be the signature of a weak overprinting event here. A shallow remanence with similar inclination is recorded in the Potluck body. Two cores in the latter unit have normal polarity and a direction similar to the Lower Miller's Dale unit intruded by this body. The Peak Forest intrusion (21) has a remanence dominated by a Recent overprint only subtracted by 500°C; an ancient normal component with a narrow range of blocking temperatures is recognised above this temperature in some samples ($D/I = 41/-18°$, $n = 6$, $R = 5.5$; see Table 3 footnote).

Remanence in the Tideswell Dale sill (site 23) is single component and magnetite-held (eg. 23–3 in Fig. 5). All cores define a well-grouped vector similar to the Waterswallows body (Table 4). The Calton Hill intrusion (site 24) possesses susceptibilities an order higher than the other bodies and remanence is dominated by large, low blocking temperature components (Table 4). Complex directional behaviour identifies a normal (NE −) component in some samples following movement from a component with SE − or NW + directions subtracted over a range of temperatures up to the Curie point of magnetite (Fig. 5). Some samples show a return to the normal direction following excursion via these directions (e.g. 24–1 in Fig. 5). Thus it is possible that slow cooling of this body has recorded a protracted geomagnetic event comprising a short term excursion beginning and ending with a normal polarity field. The paragenesis of this body is unclear. It has generally been accepted as an intrusion although recently Macdonald et al. (1984) interpret it as entirely extrusive in nature, related to a volcanic vent and separated from the Upper Miller's Dale lava by 15 m of limestone. The Miller's Dale unit has reversed polarity opposite to the only polarity recognised in the Calton Hill intrusion. Furthermore the Calton Hill rocks have high remanent intensities (Table 4) and large, low blocking temperature components unlike most of the other lava units. Hence the two are not directly related and palaeomagnetic evidence is consistent with the view that the Calton Hill body cooled and acquired its magnetisation somewhat later than the Miller's Dale unit during the normal field polarity recognised in the overlying Lees Bottom lava. The peculiar magnetic characteristics at Calton Hill may result from extensive alteration promoted here by percolating ground waters and hydrothermal solution (Macdonald et al., 1984).

The two intrusions in the SE of the Derbyshire Dome exhibit contrasting directions to the remainder. At site 25 in the Bonsall body single SE directed components are resident in magnetite with somewhat lower blocking temperatures (e.g. 25–4, Fig. 5) than intrusions to the NW. The carrier here is most probably a Ti-rich magnetite. The Ible intrusion (site 26) is also characterised by SW − directions resident in components with lower blocking temperatures than elsewhere (e.g. 26–3 and 26–8 in Fig. 5) with the same direction resid-

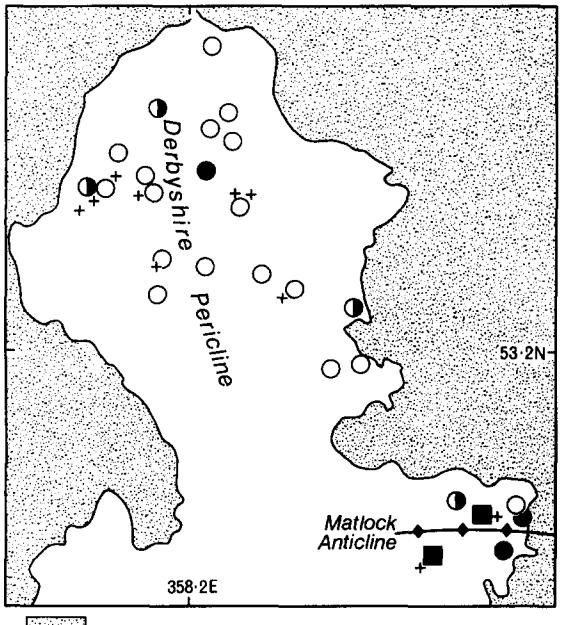



Fig. 6. Inferred nature of the characteristic remanences in the Derbyshire lavas and intrusions with major structural features of this area. Crosses refer to locations of palaeomagnetic sampling sites of earlier studies subjected to cleaning treatment and noted in the text (Section 3.3) and Table 1.

ing in two phases (see $J/J_0$ plot for sample 26–3 in Fig. 5). No sign of partial overprinting was identified in either of these bodies. They are therefore either completely overprinted or of younger age than the intrusions to the NW. The latter explanation is compatible with the possible deuteric origin of the remanence at site 26.

These two bodies are located on either side of the Matlock anticline, an east–west trending structure superimposed on the SE perimeter of the Derbyshire pericline (Fig. 6). There is marked improvement in the precision parameter $k$ from 11.1 to 64.0 when adjusted for tilt of the limbs of this structure (Table 4). The sample size is too small to evaluate the significance of this observation but suggests that remanence in these bodies (although younger than in intrusions to the NW) nevertheless predates folding of the Matlock anticline in Variscan times.

The remaining bodies have reversed polarity at Waterswallows and Tideswelldale and normal polarity at Calton Hill. The ancient remanence in the Peak Forest body is not considered to be well enough defined for inclusion in the group mean and the nearby Potluck body yields an anomalous direction not correlating with the Carboniferous or younger field. The overall mean of the remaining sites is comparable with the primary axis in the lavas. The precision improves, but not significantly so, when adjusted for the tilt of the intruded rocks (Table 4) suggesting (like the lavas) that this remanence predates folding.

Also listed in Table 4 is a mean derived from the seven components defining a WNW + /SSE − axis recognised in both lavas and intrusions. This component is not appreciably influenced by the tilt adjustment and remains enigmatic.

Although earlier studies of the Derbyshire volcanics (see locations in Fig. 6 and summary in Table 1) were based on fewer samples and a.f. demagnetisation (with the exception of some thermal demagnetisation by Titman (1971)) and components of remanence are correspondingly less well defined, they confirm the integrity of the remanence in individual bodies. Both Titman (1971) and Moyse (1973) found uniform normal polarity at sites in the Lower Miller's Dale unit and Moyse found reversed polarity in the Upper Miller's Dale unit. Titman recognised normal polarity in the Shacklow Wood lava (then interpreted as a sill) and reversed polarity in the Waterswallows intrusion. Titman also observed anomalous inclinations of remanence in the Bonsall and Ible bodies and inferred a Triassic age of magnetisation. Comparison with the palaeofield vector predicted from the APW of Eurasia now favours a somewhat older (mid-Permian) age (Fig. 4). Anomalous directions according with the WNW + /SSE − axis of Table 4 were recognised by Titman (1971) and Rolph (1984).

### *3.4. Magnetostratigraphy*

Stratigraphic data provide the basis for erecting a geomagnetic polarity time scale for Late Asbian and Brigantian Stages in the upper part of the Visean epoch (Fig. 5). The N → R reversal identified in the Miller's Dale units occurs at the boundary between these stages. A short N → R → N transition is recognised from the Lees Botton and Shacklow Wood lavas and is supported by the interpretation by MacDonald et al. (1984) of the geologic relationships at Calton Hill.

The succeeding Lower Matlock unit is reversed according to the evidence from site 12 although the inclination here is anomalously shallow. Site 13 is either comprehensively remagnetised or incorrectly mapped as a lava. The latter possibility would link it to the nearby Ible and Bonsall bodies with comparable magnetisations; this is suggested by the massively jointed outcrop and absence of vesicularity. Rolph (1984) also identified reversed polarity and a similar direction of remanence at this site.

Whilst the Conksbury Bridge unit is clearly reversed (sites 15 and 16), the polarity of the Upper Matlock unit is not readily discerned. We can conclude however that this unit is not correlative with the Conksbury Bridge unit (Walters and Ineson, 1981) and that the latter is probably older. Site 18 lies in a separate flow unit a few metres above site 17. A transitional field direction recognised at site 17 ($D/I = 317/21°$) is not widely removed from a direction resolved by Rolph (1984, site BW1) from a.f. demagnetisation of the same unit ($D/I = 343/39°$). Recognition elsewhere in

TABLE 5

Carboniferous to Permian palaeomagnetic poles from the British Isles

| No. | Rock formation | (South) Pole position | | Age (Ma) | Reference |
|---|---|---|---|---|---|
| | | °E | °S | | |
| *(i) Lower Carboniferous* | | | | | |
| 1 | Derbyshire lavas | 336 | 14 | about 335 | This study |
| 2 | Derbyshire intrusions | 329 | 12 | | This study |
| 3 | Kinghorn lavas | 343 | 18 | | Wilson and Everitt (1963) |
| 4 | Kinghorn–Burnitsland lavas | 332 | 14 | | Torsvik et al. (1989) |
| 5 | Largs lavas | 322 | 14 | | Torsvik et al. (1989) |
| *(ii) Upper Carboniferous-Permian* | | | | | |
| 6 | Derbyshire lavas, overprint | 313 | 42 | | This study |
| 7 | Derbyshire, SE intrusions | 304 | 38 | 309 | This study |
| 8 | Dykes, NW Highlands | 355 | 35 | 320 | Esang and Piper (1984) |
| 9 | Sill, Queensferry | 354 | 38 | | Torsvik et al. (1989) |
| 10 | Dyke, Peterhead | 342 | 41 | | Torsvik (1985) |
| 11 | Lava overprints and dykes, largs | 343 | 43 | | Torsvik et al. (1989) |
| 12 | Whin sill | 339 | 44 | 281, 295 | Storetvedt and Gidskehaug (1969) |
| 13 | Exeter lavas | 345 | 46 | 281, 279 | Cornwell (1967) |
| 14 | Wackerfield dyke | 349 | 49 | 303 | Tarling and Mitchell (1973) |
| 15 | Dykes, Outer Hebrides | 347 | 47 | | Piper (1991) |

this area (Fig. 4) implies that it is of real geomagnetic and/or geological significance. The uppermost site 18 has normal polarity but a direction only compatible with a much younger palaeofield direction (Fig. 4). Four, and probably five, inversions of the geomagnetic field are therefore identified between Upper Asbian and Brigantian times (Fig. 7). This time period is at about 333 Ma according to the time scale of Harland et al. (1982). Asbian and Brigantian times comprise two of the five Stages of the Visean epoch between about 352 and 333 Ma; hence the time interval represented by this polarity stratigraphy is likely to be about 4 Ma.

Palmer et al. (1986) discuss the magnetostratigraphy of Carboniferous times and summarise evidence from four separate sections of carbonate lithology spaning Asbian and Brigantian times after Perry (1979). Unfortunately, age of the remanence in these rocks is unclear (Section 4) and only a normal event at the Asbian/Brigantian boundary has emerged. At least two normal events are now recognised in the Brigantian Stage in place of uniform reversed polarity shown by these authors.

## 4. Discussion

The majority of results from this study define a NE − /SW + axis which (i) is of dual polarity, (ii) resides in magnetite, (iii) appears to predate intra-Carboniferous folding and (iv) does not correlate with the post-310 Ma palaeofield vector. It is therefore interpreted to be a primary Lower Carboniferous pole position and is identical, within error limits, to poles derived from the Lower Carboniferous volcanics of the Midland Valley of Scotland (Table 5 and Fig. 8(a)). The latter studies (based formerly on the work of Wilson and Everitt (1963) and subsequently confirmed by Torsvik et al. (1989)) do not yield a significant fold test, but the presence of reversals and the integrity of polarity within individual lava units strongly support a primary origin (Torsvik et al., 1989). Collectively, these results provide strong confirmation of the Lower Carboniferous pole position for Europe. They show that later tectonic deformations between central England and the Midland Valley of Scotland during the Variscan episode (see for example Leeder, 1982) are below the resolution of palaeomagnetism.

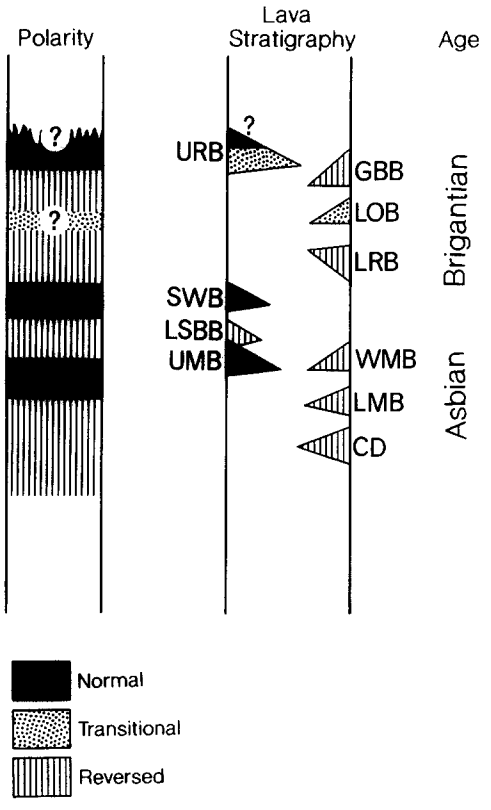


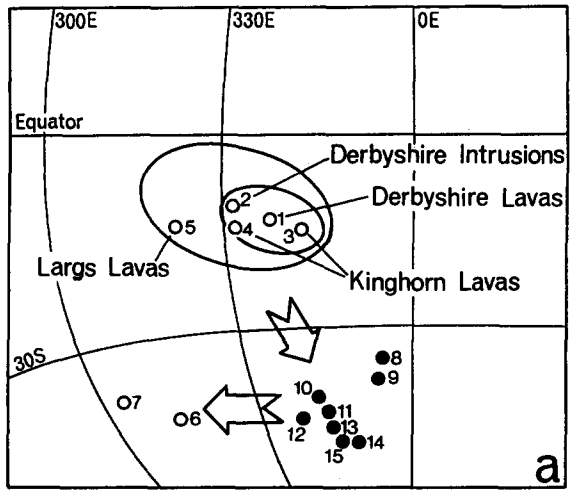

Fig. 7. Magnetostratigraphy of the upper Asbian and Brigantian Stages (comprising the last part of the Visean times) deduced from palaeomagnetic results from the lavas in the Derbyshire succession. LSBB is the Lees Bottom unit; other units are referenced in the caption to Fig. 1.

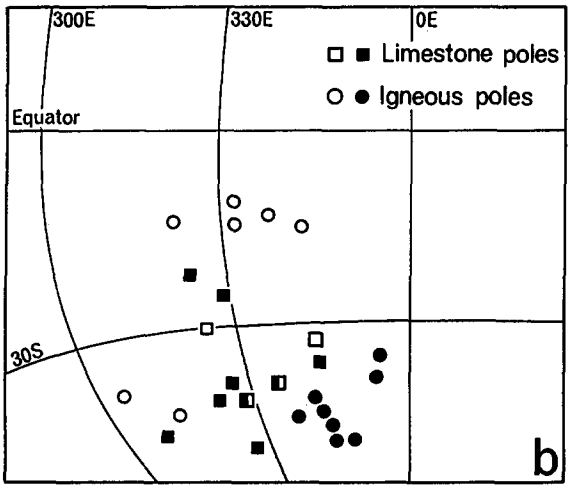
The poles are removed by some 30° from a group of poles (section (ii) in Table 5) less well defined by palaeomagnetic field tests and assigned to Upper Carboniferous and Permian times. This part of the APW path is well represented in continental Europe (eg. Halvorsen et al., 1989). Result 8 is assigned to the Lower–Upper Carboniferous boundary and the bulk of the implied APW movement therefore appears to have taken place during Lower Carboniferous times between about 335 and 320 Ma. The second group of poles is distributed over little more than 10° and suggests that only small polar movement occurred during Upper Carboniferous and early Permian times (Fig. 8(a)).

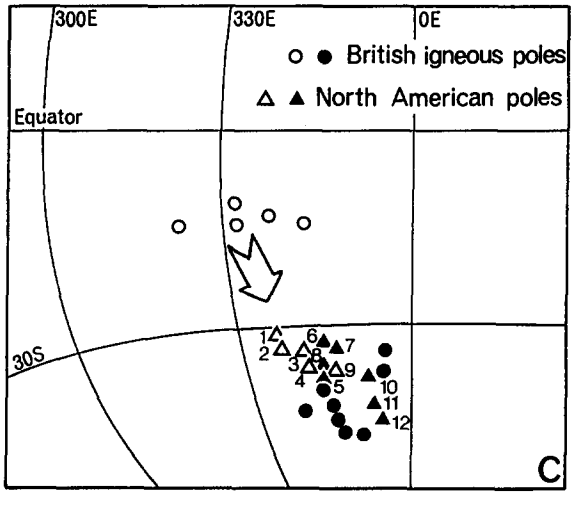
The Derbyshire volcanics are interleaved with thick and widespread shelf limestones. The palaeomagnetism of these limestones has been extensively studied in the Pennines of England immediately to the north of this study area (Turner and Tarling, 1975; Turner et al., 1979, Addison et al., 1985). Directions of characteristic remanence yield pole positions which plot typically between 20° and 30° to the south of the Lower Carboniferous pole position recognised from the igneous formations of this age (Fig. 8 (b)), and mostly to the west of the Upper Carboniferous–Permian poles of division (ii) in Table 5. An inclination error is not an adequate explanation for this discrepancy because the difference in palaeomagnetic direction is one of declination as well as inclination (S–SSW compared with SW). Hence we conclude that the limestones are dominated by an overprinted remanence of post Lower Carboniferous age. They are therefore unreliable recorders of Carboniferous magnetostratigraphy and tectonics. This conclusion is supported by their polarity: some 90 % of the limestones have reversed magnetisation consistent with remanence acquisition during the Carboniferous–Permian reversed superchron (see Harland et al., 1982; Piper, 1987). In contrast 30 % of the volcanics which have not been overprinted have normal polarity.

The directional discrepancy is also evident in the Derbyshire region. Figure 9 is an example of remanence in a limestone beneath the Upper Miller's Dale unit. The characteristic remanence is southerly shallow in contrast to the SW + component resolved from the lavas (see Fig. 2). Similar demagnetisation results are illustrated by Addison et al. (1985) from the northern Pennines.

Curiously, the pole plotting closest to the Lower Carboniferous position is derived from the Draughton Limestone (324°E, 21°S) which the authors recognised as secondary and attributed to superimposed normal and reversed components (Turner et al., 1979). These limestones are strongly silicified and other results in the literature from strongly silicified units (see for example, Addison et al., 1985, fig. 8) show that these rocks have shallow positive SSW directed magnetisations closer to the field direction now recognised to be characteristic of Lower Carboniferous times and contrasting with the SSW negative directions in unsilicified limestones. Hence it would appear that




silicification during diagenesis has tended to shield these sediments from the effects of subsequent overprinting. On the basis of the derived pole positions, the Lower Carboniferous limestones appear to have a remanence which mostly postdates early Permian times (c.f. poles 8 and 12–14 in Table 5 and Fig. 8(a)). They are distributed towards the pole derived from overprints in the Derbyshire Volcanics (6) and the probable later intrusive event in SE Derbyshire (7). These characteristic directions of remanence are recognised in both dolomitised and undolomitised limestones of the Pendleside Formation in the northern Pennines (Addison et al., 1985). Hence we interpret them as overprints imparted during a post-early Permian heating event of regional extent which is expressed locally in SE Derbyshire as an igneous event and recognised by repeated hydrothermal activity between 270 and 235 Ma (Ineson and Mitchell, 1973).

Palaeomagnetic poles from Lower Carboniferous limestones of Ireland plot in this same location (see Piper, 1988 for a summary of data) and we infer that the remanence record here has a similar origin. A single a.f. demagnetised result from Lower Carboniferous volcanics in the Limerick Basin (Mulder, 1972) defines a comparable pole position (329°E, 39°S); it is unsupported by field tests and seems likely to record the same overprinting episode.

The Carboniferous palaeomagnetic record in North America has been the subject of extensive

Fig. 8. (a) Palaeomagnetic poles derived from a.f. and/or thermally treated igneous rocks of Lower Carboniferous to early Permian age from Britain. Open symbols are Lower Carboniferous poles with 95 % confidence ovals derived by this investigation. Poles are referenced in Table 5. (b) Palaeomagnetic poles derived from Lower Carboniferous limestones in Britain compared with pole positions from igneous units in (a). Reversed polarities in the limestones are shown as the closed fraction of the symbol and normal polarities as the open fraction. (c) Palaeomagnetic poles from Lower Carboniferous rocks of North America corrected for later drift according to the reconstruction of Bullard et al. (1965). Open triangles are mixed polarity results and closed triangles are exclusively of reversed polarity. The numbers reference the poles after Roy and Morris (1983) as follows: 1 (M); 2 (S); 3 (E); 4 (MS); 5 (H); 6(RV); 7 (AG); 8 (C); 9 (R); 10 (B); 11 (T); 12 (PE).

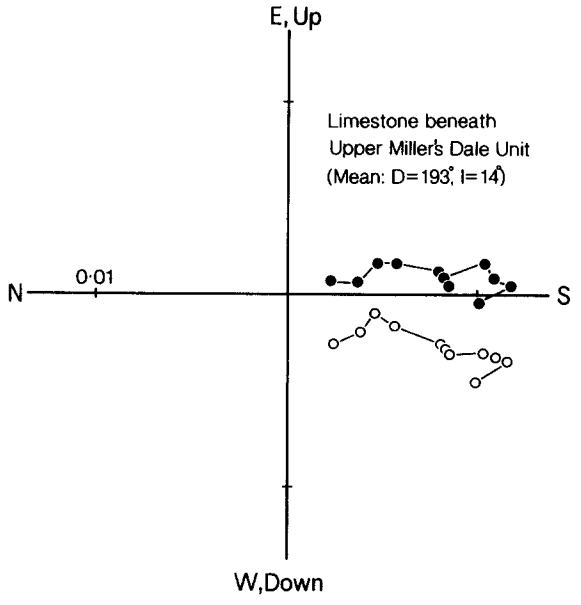



Fig. 9. Example of remanence residing in limestone beneath the Upper Miller's Dale lava near site 7. Demagnetisation steps are 5 mT and symbols are as for Fig. 2.

discussion in the literature. We note one implication here in the context of the new British data. Roy and Morris (1983) recognise a NW–SE trend in (reversed) Carboniferous poles from the eastern margin of North America. They observe that poles plotting on the earlier part of the APW trend have dual polarities and are therefore more likely to incorporate primary magnetisations. Poles plotting on the latter part of the trend are exclusively of reversed polarity. There is therefore suspicion that they record overprints acquired during the Carboniferous–Permian superchron. The boundary between these groups of poles is an 'horizon marker' defining the beginning of this superchron. When these poles are rotated into the coordinate frame of the British poles employing the conventional reconstruction of the North Atlantic continents (Bullard et al., 1965; Van der Voo, 1990), it is observed (Fig. 8(c)) that the swathe of Roy and Morris (1983) records the latter part of the Lower Carboniferous APW motion recognised by this analysis. The reversely-magnetised segment of the path corresponds to the group of Upper Carboniferous–Permian poles from igneous units identified in Britain (Table 5(ii)). It links these poles collectively to magnetisation during the Carboniferous–Permian superchron.

The North American poles are derived predominantly from sedimentary units and Roy and Morris (1983) note that even magnetisations which are demonstrably prefolding in origin can be secondary. This may explain why only the latter part of the Carboniferous APW trend recognised in Britain is currently recorded by the North American data (Fig. 8(c)).

## Acknowledgements

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